

SMART WASTE CONNECT

AI-Driven
Waste
Management
System for
Pokhara
Proposed
Project
Presentation

Framework:
Django 6.0.4

Problem Statement

- **Overflowing Containers:** Public garbage bins in Pokhara often overflow, causing pollution and health hazards.
- **Inefficient Collection:** Waste collection trucks follow fixed routes regardless of whether bins are full or empty.
- **Data Verifiability:** Citizen reporting systems often suffer from "spam" or fake photo uploads.
- **Lack of Real-time Tracking:** Citizens cannot see when a reported issue will be addressed.

Proposed Solution

Citizen Crowdsourcing

- Public can snap photos of full green containers.
- Location is automatically captured via GPS metadata.
- Login-free reporting to encourage participation.

AI Verification (ML)

- Trained model to recognize specific "Green Containers".
- Filters out spam/fake images instantly.
- Ensures only valid reports reach the municipality.

Tracking & Security Mechanism

Daraz-style Tracking

Status-based tracking system for better UX:

- **Reported:** User submits photo.
- **Verified:** AI confirms it's a bin.
- **Assigned:** Closest driver receives alert.
- **Completed:** Bin emptied.

Security Protocols

- **Manual Registration:** Drivers/Admin accounts created only by Municipality to prevent breaches.
- **Account Lock:** Login required only for tracking features to protect user data.
- **EXIF Validation:** Checks if photo was taken at the reported location.

Technical Architecture

Backend: Django 6.0.4 (Robust and Secure)

Database: PostgreSQL with PostGIS (For spatial/map data)

Frontend: Responsive Templates + Leaflet.js (Map Integration)

ML Engine: TensorFlow/Keras (MobileNetV2 for Image Classification)

Python

GeoDjango

Bootstrap

OpenStreetMap

SDLC (Agile Methodology)

Week	Focus Area	Key Deliverables
Week 1	Planning & Design	DB Schema, UI Mockups, Env Setup
Week 2	Core Development	Auth, Photo Uploads, Admin Dashboard
Week 3	AI & Maps	ML Model Integration, Leaflet Maps
Week 4	Testing & Deploy	Bug Fixes, Presentation for Municipality

Conclusion & Future Scope

- **Scalability:** Can be expanded to all wards of Pokhara Municipality.
- **Optimized Logistics:** Reduces fuel consumption for garbage trucks by 30%.
- **Gamification:** Reward points for citizens who report frequently.
- **Smart Sensors:** Future integration with IoT ultrasonic sensors inside bins.

Clean Pokhara, Green Pokhara.